

Project Management – Basic Concepts

Topic # 1

Chapter 1 – PMBOK, Schwalbe

Chapter 1, 2 – Reference Material

## Project Attributes & Project Manager

- Project has a unique purpose
- Project is temporary
- A project is delivered using progressive elaboration
- A project requires resources, often from various areas
- A project should have primary customer or sponsor
- A project involves uncertainty

*An effective project manager is crucial to a project's success. Project managers work with the project sponsors, the project team and the other people involved to meet project goals*

## What are Projects?




- A **project** is a temporary endeavor undertaken to create a unique product, service, or result.
- A project can create:
  - A product that can be either a component of another item, an enhancement of an item, or an end item in itself;
  - A service or a capability to perform a service
  - An improvement in the existing product or service lines ; or
  - A result, such as an outcome or document

## Project Types

- Software development
- >Software integration
- Database deployments
- Product development
- Product Innovation/Specialization
- Acquisitions
- Process Improvement




IT Project

1. A team of students creates a smartphone application and sells it online.
2. A company develops a driverless car.
3. A small software development team adds a new feature to an internal software application for the finance department.
4. A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus.
5. A company develops a new system to increase sales force productivity and customer relationship management that will work on various laptops, smartphones, and tablets.
6. Internet of things: Internet usage will expand as sensors are added to physical items that are connected to the Internet. For example, Near Field Communication allows users to make payments, board airplanes, and perform other tasks by placing their phones in front of a reader
7. A large group of volunteers from organizations throughout the world develops standards for environmentally friendly or green IT.
8. Government regulations require new reporting of commercial business data for a manufacturing company.




| Project | Operations |
|---|---|
| Performed by people | Performed by people |
| Constraint by limited resources | Constraint by limited resources |
| Planned, executed, and controlled | Planned, executed, and controlled |
| *Projects are temporary and unique* | *Operations are ongoing and repetitive* |
| *Projects are different from operations in that they end when their objectives have been reached or the project has been terminated* | *Ongoing operations have the opposite characteristics of projects. It is the work done in the organization to sustain the business* |




**Project Triple Constraints**

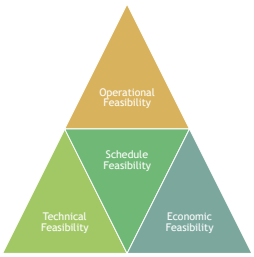


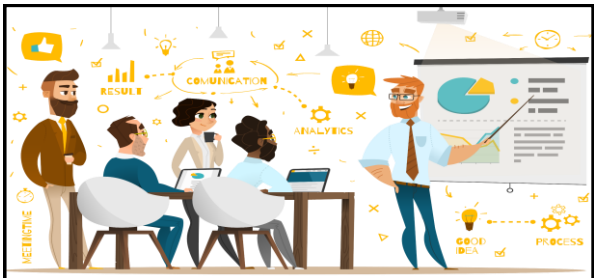

| Constraint Change | Required Adjustment Alternatives (One or Combination of Both) | |
|---|---|---|
| Shorter Time | Higher Cost | Reduced Quality or Narrowed Scope |
| Reduced Cost | More Time | Reduced Quality or Narrowed Scope |
| Higher Quality or Increased Scope | More Time | Higher Cost |

Increased **Scope** = increased time + increased cost

- Tight **Time**= increased costs + reduced scope
- Tight **Budget**= increased time + reduced scope.

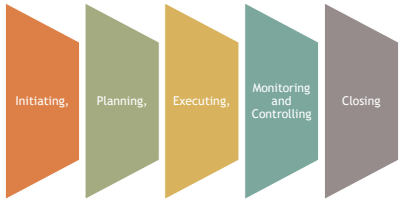
**Feasibility Study**



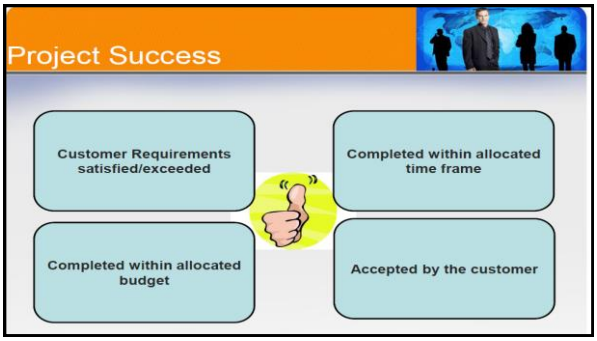




**Project Management**

- Project management is the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.
- These five Process Groups are integrated and applied by PM:




- Managing a project typically includes, but is not limited to:
  - Identifying requirements;
  - Addressing the various needs, concerns, and expectations of the stakeholders in planning and executing the project;
  - Setting up, maintaining, and carrying out communications among stakeholders that are active, effective and collaborative in nature;
  - Managing stakeholders towards meeting project requirements and creating project deliverables;
- Balancing the competing project constraints, which include, but are not limited to:
  1. Scope
  2. Quality
  3. Schedule
  4. Budget
  5. Resources
  6. Risks.



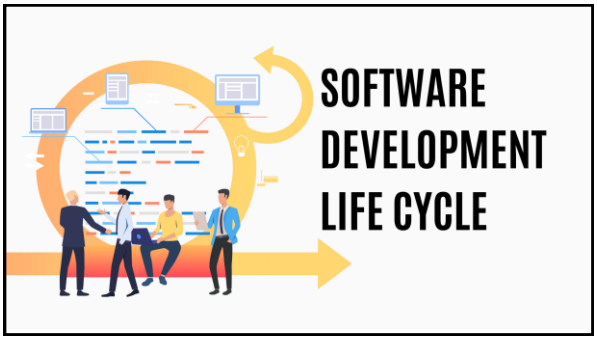






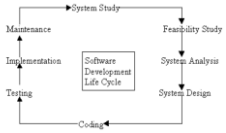
**Project Failure Causes**

| | | | |
|---|---|---|---|
| Failure to establish upper-management commitment to the project | Lack of organization's commitment to the system development methodology | Taking shortcuts through or around the system development methodology | Poor expectations management |
| Premature commitment to a fixed budget and schedule | Poor estimating techniques | Over-optimism | Inadequate people management skills |
| Failure to adapt to business change | Insufficient resources | Failure to "manage to the plan" | Communication Gaps |




## Software Development Life Cycle (SDLC)

- Software Development Life Cycle (SDLC) is a process used by the software industry to design, develop and test high quality software.
- It consists of a detailed plan describing how to develop and maintain software.
- SDLC consists of many activities/ phases.
- Following are the major phases of SDLC.
  - System study
  - Feasibility study
  - System analysis
  - System design
  - Coding
  - Testing
  - Built release
  - Maintenance




## What are guidelines for system development?

Arrange tasks into phases (groups of activities)

Involve users (anyone for whom system is being built)

Develop clearly defined standards (procedures company expects employees to follow)




## Project Stakeholders

- Project stakeholders are individuals and organizations that are actively involved in the project or whose interest May be affected as a result of project execution or project completion
- Important to identify the stakeholders:
  - Who the stakeholders are
  - What role they will play
- **Two groups of stakeholders**
  - **Internal stakeholders:** Company owner, shareholders, project managers, technical managers, team leaders,developers, and other employees.
  - **External stakeholders**: Government, military, customers, suppliers, competitors etc.

## The Players of the System – Stakeholders

| | |
|---|---|
| System owners | • An information system's sponsor and executive advocate, usually responsible for funding the project of developing, operating, and maintaining the information system |
| System User | • A "customer" who will use or is affected by an information system on a regular basis - capturing, validating, entering, responding to, storing, and exchanging data and information<br>• Internal users<br>• External users<br>• Direct users & Indirect users<br>• Non human users |
| System Designer | • A technical specialist who translates system users' business requirements and constraints into technical solution. She or he designs the computer databases, inputs, outputs, screens, networks, and software that will meet the system users' requirements |
| System Builder | • A technical specialist who constructs information systems and components based on the design specifications generated by the system designers |
| Software Quality Assurance People | |
| Project Manager | • An experienced professional who accepts responsibility for planning, monitoring, and controlling projects with respect to schedule, budget, deliverables, customer satisfaction, technical standards, and system quality. |
| External Service Provider (ESP) | • A systems analyst, system designer, or system builder who sells his or her expertise and experience to other businesses to help those businesses purchase, develop, or integrate their information systems solutions; may be affiliated with a consulting or services organization. |
| System Analyst | • A specialist who studies the problems and needs of an organization to determine how people, data, processes, and information technology can best accomplish improvements for the business. |

## Project & Process Management

**Project**

- a [temporary] sequence of unique, complex, and connected activities having one goal or purpose and that must be completed by specific time, within budget, and according to specification

**Project management**

- The process of scoping, planning, staffing, organizing, directing, and controlling the development of an acceptable system at a minimum cost within a specified time frame

**Process management**

- the activity of documenting, managing, and continually improving the process of systems development.

## Role of Project Managers

The project manager is the person assigned by the performing organization to lead the team that is responsible for achieving the project objectives.

In general, project managers have the responsibility to satisfy the needs: task needs, team needs, and individual needs.

Build customer-company alliance.

Link between company strategy- project.

## Project Manager Competencies

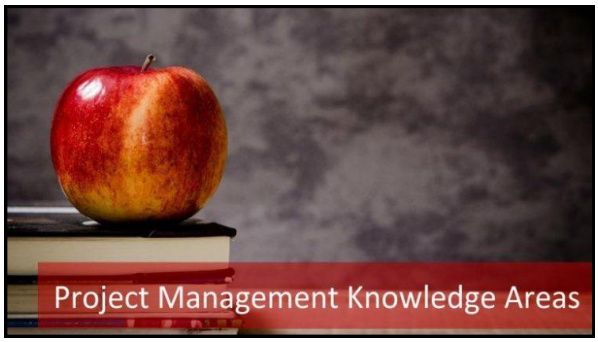
- Effective project management requires that the project manager possess the following competencies:

| Knowledge | Performance | Personal |
|---|---|---|
| • Refers to what the project manager knows about project management | • Refers to what the project manager is able to do or accomplish while applying his or her project management knowledge | • Refers to how the project manager behaves when performing the project or related activity. Personal effectiveness encompasses attitudes, core personality characteristics, and leadership, which provides the ability to guide the project team while achieving project objectives and balancing the project constraints. |

## Project Manager – Interpersonal Skills

- Project managers accomplish work through the project team and other stakeholders. A proper balance of ethical, interpersonal & conceptual skills is needed.

Leadership,
Team building,
Motivation,
Communication,
Influencing,
Decision making,
Political and cultural awareness,
Negotiation,
Trust building,
Conflict management, and
Coaching




Project Integration Management
Project Scope Management
Project Time Management
Project Cost Management
Project Quality Management
Project Human Resource Management
Project Communication Management
Project Risk Management
Project Procurement Management
Project Stakeholder Management

## Project Integration Management

This knowledge area focuses on project plan develop and execution

•It includes:

-Creation and approval of project plan

-Executing the project plan

-Managing , controlling and documenting changes to the project plan

## Project Scope Management

Primarily it is the definition and control of what ***IS*** and ***IS NOT*** included in the project.

•Project Scope Management is the process to ensure that the project is inclusive of all the work required, and only the work required, for successful completion.

•Project Scope Management plan includes:

•collect requirements

•Planning the project scope

•Defining the exact project scope

•Create WBS

•Verifying the project scope

•Controlling the project scope

## Project Time Management

- This knowledge area cover activities their characteristics and how they fit into the project schedule

•It includes:

-Scheduling Activities

-Sequence Activities

-Estimate Activity Resources

-Estimate Activity Duration

-Develop Schedule

-Control Schedule

## Project Cost Management

- This process is required to ensure the project is completed within the approved budget
  - **Resources**
  - people equipment
  - materials
  - **Quantities**
- This knowledge area is concerns with planning, estimating, budgeting and control of cost.

•It includes

-Estimate Cost

-Creating the project budget using project management cost controlled techniques

Control cost

## Project Quality Management

1. **User-Based:** Fitness for use, meeting customer expectations.
2. **Manufacturing-Based: Conforming** to design, specifications, or requirements. Having no defects. Quality means conformance to process standards.
3. **Product-Based:** Quality can be appreciated by measuring the inherent characteristics of the product which will result in improved external product behavior
4. **Value-Based:** The product is the best combination of price and features.
5. **Transcendent:** It is generally associated with some intangible properties that delight users.

## Project Quality Management

- Quality Management is the process that ensure the project will meet the needs
- "conformance to requirements" -**Crosby**

  "fitness for use" -**Juran**
- "the totality of characteristics of an entity that bear on its ability to satisfy stated and implied need" -**ISO 8402:1994**
- It deals with the quality planning, assurance and control.
- It includes:
  -Planning for project quality
  -adhere to quality assurance
  -Enforcing set quality control systems

## Project Human Resource Management

- It focuses on organizational planning, staff acquisition and team development
- It includes:
  - Developing a project organizational structure or HR Plan
  - Acquire project team
  - Developing the project team
  - Manage team
  - Resolve Conflicts (force, compromise, smooth, confront, withdraw, collaborate)

## Project Communication Management

- This process is necessary to ensure timely and appropriate generation, collection, dissemination, and storage of project information
- **Communications planning:** Determining the needs (who needs what information, when they need it, and how it will be delivered)
- **Information Distribution:** Defining who and how information will flow to the project stakeholders and the frequency
- **Performance Reporting:** Providing project performance updates via status reporting.

## Project Risk Management

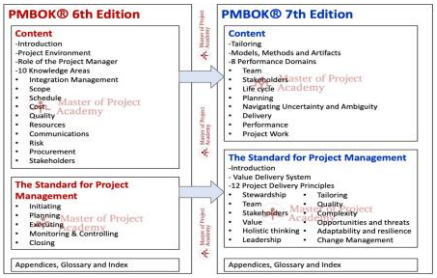
- Find Potential Negative Impact on the project
- This knowledge area focuses on risk planning, analysis, monitoring and control.
- Risk is an art and science of identifying, analyzing responding to risk throughout the life of project.
- Risk is an uncertainty that can have a negative or positive effect on meeting project objectives
- It includes:
  - Planning for risk management
  - Identifying risk
  - Using qualitative risk analysis
  - Using quantitative risk analysis
  - Creating project risk plans
  - Actively monitoring and reacting to project risk

## Project Procurement Management

- This knowledge area involve solicitation , contract administration and contract closeout
- It includes :
  - Planning for project procurement
  - Planning for solicitation
  - Management Project solicitation
  - Selecting vendors
  - Managing and creating procurement documents
  - Administrating and closing project documents

## Project Stakeholder Management

- **Stakeholder management:**
  - Includes identifying and analyzing stakeholder needs while managing and controlling their engagement throughout the life of the project.



# END OF TOPIC 1

- COMING UP!!!!!!
- Project Lifecycle
- Project Teams
- Project Initiation & Integration

